

Hunt the Wumpus: Assignment 4

In this assignment we're adding a user interface: a console application that accepts user inputs and calls into the game to make moves. We also need to add printouts of the text expected when different events happen in the game. A new class, `UserNotification`, needs to be added. Define a set of possible events with an enumeration, and then support registering callback functions for when those events trigger. See the specification for the set of events.

Insert triggering the events at appropriate points in the code. E.g. when the Hunter enters the cave, report the caves connected, and report the denizens of neighboring caves, or when the bat picks up the Hunter. In the unit tests, verify that the appropriate registered callback is triggered for a game operation. The provided code is untested (you're given the Assignment 3 Solution with some small modifications, which don't even compile for this code).

When this assignment is complete, you should be able to play Hunt the Wumpus by entering "move <cave number>" or "shoot <cave number> <cave number> <cave number>" (for arrow flight paths).

Criteria	Ratings	Points
User Notification class	Class supports registering <code>std::function</code> callbacks.	5
User Notifications: Main	The main program registers callbacks and makes printouts to <code>std::cout</code> that indicate the event happened.	5
User Notifications: Cave	A cave reports its cave number and the set of tunnels when the Hunter enters and isn't affected by a Denizen. A unit test verifies the right numbers are present.	2
User Notifications: Cave Denizens	A cave reports the denizens contained (e.g. "I smell a Wumpus!" as in the specification) when the Hunter enters and isn't affected by a Denizen. A unit test verifies the callback is triggered.	2
User Notifications: Pit	The pit reports that the Hunter fell in. A unit test verifies that the callback is triggered.	2
User Notifications: Bat	The bat reports that it grabbed the Hunter. A unit test verifies that the callback is triggered.	2
User Notifications: Wumpus	The Wumpus reports when it is bumped into by the Hunter, or eats the Hunter when bumped into, or is shot by an arrow. Unit tests verify all behaviors.	2
User Notifications: Hunter	The Hunter reports when it's hit by an Arrow or eaten by the Wumpus entering its Cave. Unit tests verify all behaviors.	2
User Notifications: Arrow	The game reports when the Hunter is out of arrows and loses. The game reports when missing with Arrows wakes up the Wumpus.	2
User Notification: Illegal Move	The processing engine reports when a move is illegal.	1

Criteria	Ratings	Points
	A unit test verifies that the callback is triggered.	